DMFC(**D**igital **M**ass **F**low **C**ontroller) & DMFM Protocol Manual

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Draft version	

Line tech

Documents revisions

Doc. Ver.	Description	dates
1	Draft	
1.1	Addition of calibration command	2014, 06,
1.2	Addition of Calibration command.	2015, 01,
1.3	Commands summary table form changed	2015, 04
1.5	Commands summary table form changed	2015, 05
	Status Code changed 0xBn -> 0xEn	2015, 10
1.6E	First English version manual	2016, 07

Firmware Revisions

F/W Ver.	Descriptions	dates
1.C	First release version	2015, 10
1.E	Cutout function bug fix	2015, 11
1F	While voltage/current calibration, the DMFC's mode changing bug fix	2015. 11
2.2	Baud rate are 9600, 19200, 38600, 576000, 115200	2015, 12
2.2A	EEPROM format updating function error fixed.	2015, 12
…	…	
2.6	Input output cut off range function error fixed and other many bugs are fixed	2016, 5

Index

1. DMFC Serial Communication basics
2. DMFC Protocols
3. User command summary
4. Appendix 1
5. Appendix 2
6. Appendix 3

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Chapter 1. DMFC digital communication basics

1.1 Basic description

Digital gas Mass Flow Controller (DMFC) and Digital Mass Flow Meter (DMFM) has RS-485 communication port and protocol for setting and reading the gas flow rates and information. It can simultaneously connect up to 32 DMFC (DMFM) at RS-485 communication line

1.2 DMFC/DMFM Specifications

- Communication Protocol: RS-485,
    - Assignable address ranges are from 1 to 254.
    - But the DMFC (DMFM) on the communication line are limited 32 each.
- Connection: DB15(Male)
- Default setting: 38400, n, 8, 1, ASCII
    - User selectable baud rates are 9600, 19200, 38400, 57600.

1.3 Connection

Use the 14 and 15 pin of D-sub connector for the RS-485 +/- .

Pin No. (DB15 Male)	Dir.	Descriptions	Pin No. (DB15 Male)	Dir.	Descriptions
1		signal common	9		Power common
2	O	voltage sig. out	10		Signal common
3	O	Alarm output	11	O	Reference 5V
4	O	Current sig. out	12	I	Valve override
5		Power	13	X	N. C.
6	X	N. C.	14	⇔	RS-485 -
7	I	Current sig in	15	⇔	RS-485 +
8	I	Voltage sig in			

Chapter 2. Protocols and command frame

Sending frame (PC to DFMC)

FORMAT	HEAD	Address	Did	Mode	Command	[Data]	Checksum	TAIL
No. char	**1**	**2**	**1**	**1**	**2**	**n**	**2**	**1**

Receiving frame (DMFC to PC)

FORMAT	HEAD	Address	Did	Command	[Data]	Status	Checksum	TAIL
No. char	**1**	**2**	**1**	**2**	**N**	**2**	**2**	**1**

Memo ::

※ The data element is used at write command and response for read command only

※ The [STATUS] element is used at response only.

2.1 HEAD (1 char.)

Direction	Normal	On error
PC => DMFC	STX(0x02)	-
DMFC => PC	ACK(0x06)	NCK(0x15)

2.2 Address (2 char.)

DMFC(DMFM) address range

- "01" ~ "FE" ("FF" is for the broadcasting only, or 1 to 1 communication)

Memo

- Only some commands can use the "FF" address for broadcasting.
  - Example. All DMFC ON/OFF, Integration Start/Stop…
  - Another application of FF address is look for the address of the one connected DMFC/DMFM.

2.3 Device ID(1 char.)

Since this protocol was created in consideration of the communication with other devices, command frames need an ID to specifying the DMFC/DMFM. The device ID of DMFC/DMFM is a letter 'D' (0x44).

2.4 Mode (1 char.)

The letter to distinguish the type of command (1char).

Mode	Spec.	Response
"R"	Read Value	Response for the command
"W"	Write Value	**No response**
"w"	Write & Read Value	Same response for the "R" commands

Attention

If there's error on writing command frame, DMFC would not send any error message. If you want to check the result of writing command, please check with the matched reading command.

2.5 Command and data

Command

Commands are consisted with two letters and optional data.

(2 letter command) + [DATA]

For the descriptions of the commands, see chapter 3.

Data

There's four data types and it's combinations. The basic four data types are :

1) A value within the range 0~255 is sent by two hexadecimal number letters. At the commands summary table, this data type is marked by 'H'.
   Ex. 1 → "01", 129 → "81"
2) The integer number within the range 0~9999, is sent by for letter leading zero string. At the commands summary table, this data type is marked by 'D'
   Ex. 1234 → "1234', 789 → 0789 …
3) The floating-point number is sent by IEEE-754 little endian 4byte as hexadecimal 8 letters string. At the commands summary table, this data type is marked by 'F'.
   Ex. 2.5 → "00002040"
4) String. At the commands summary table, this data type is marked by 'S'.
   Ex. "TAG-NAME:123456:"

Memo ::

Most read commands send command only. Some write commands not require optional data.

2.6 Data type specifiers at command tables

Data format specifiers

specifier	Descriptions
H	Two letter hexadecimal number Ex. 00, 01, 02
D	Leading zero four letter decimal number. Ex. 1234, 0123
F	Sends IEEE-754 little endian 4 byte floating point number as 8 letters hexadecimal Ex. 2.5 → "00002040"
S	String. Letters and numbers. Ex. TAG-NAME:123456:
--	No read or write command at this command number

Examples

Basic types

2H/-- Read only command and DMFC/DMFM sends two letter data.

8F Read and write command with 8 characters float.

Combination type

8H Modified for read/write four 2 letters hexadecimal numbers.

16S 16 characters string

8S8F 8 characters and 8 character float number.

2.6 Status (2 char.)

If user send frame is correct, DMFC responding frame has following status values.

Value (hex)	Description
0x00	Ready , OK
0xE0	Not OK
0xE1	Not a command
0xE2	No right
0xE4	reserved
0xE8	reserved

2.7 Checksum (2 char.)

The checksum value is the XORing result from address to status.

Ex. Read current flow rate of address 9 DMFC/DMFM

Sending frame: <STX> + "09"+"D"+"R" + "01"+ "1E" + <ETX>

Checksum: 0x30 **xor** 0x39 **xor** 0x44 **xor** 0x52 **xor** 0x30 **xor** 0x31 = 0x1E

	Address		Dev. ID	mode	command		Check sum		TAIL
<STX>	0	9	D	R	0	1	1	E	<ETX>
0x02	**0x30**	**0x39**	**0x44**	**0x52**	**0x30**	**0x31**	0x31	0x43	0x03
	0x30 ^ 0x39 ^ 0x44 ^ 0x52 ^ 0x30 ^ 0x31 = 0x1E								

2.8 TAIL (1 char.)

The 'ETX(0x03)' character notifies the end of command frame.

2.9 Programming Example

Example 1. Flow set-point read/write command(06). Data format is 8F, DMFC address is 10(0x0A)

"R": Read

Sending : <STX>+"0A"+"D"+"**R**"+"09"+"6E"+<ETX>

<STX>: 0x02

"0A": Address

"D": Device ID(for DMFC/DMFM)

"R": Read command

"09": Set-point R/W Command No.

"6E": Checksum value. Hex number string of XOR ing from address before checksum

0x30("0") xor 0x41("A") xor 0x44("D") xor 0x52("R") xor 0x30("0") xor 0x39("9")

⇒ 0x6E ⇒ "6E"

<ETX>: 0x03

Response : <ACK>+"0A"+"D"+09"+"00004842"+"00"+"36"+<ETX>

<ACK>: 0x06

"00004842": 0x00004842=>50.0 (Little endian hex number string)

"00": Status

<ETX>: 0x03

"W": Write

Sending: <STX>+"0A"+"D"+"**W**"+09"+"00005C42"+"1B"+<ETX>

"00005C42": 0x00005C42=>55.0 (4byte 실수=>Little endian 16 진수 String)

Response : no response.

"w": Write

Sending : <STX>+"0A"+"D"+"**w**"+09"+"00005C42"+"3B"+<ETX>

Response : <ACK>+"0A"+"D"+09"+"00005C42"+"00"+"4C"+<ETX>

Same as "R" command

Example 2. Set-point source read/write command(01). The data type is 2H. DMFC address is 10(0x0A).

"R": Read

Sending : <STX>+"0A"+"D"+"R"+"**01**"+"66"+<ETX>

"01": Set-point source read command number.

Response : <ACK>+"0A"+"D"+01"+"**01**"+"00"+"35"+<ETX>

"01": Current flow set-point is set by digital command(digital mode).

(if the data is "00", current mode is analog mode.)

"W": Write

Sending : <STX>+"0A"+"D"+"W"+"01"+"**00**"+"63"+<ETX>

"00": 0x00, Set analog mode.

Response : No response

"w": Write

Sending rame: <STX>+"0A"+"D"+"w"+"01"+"**00**"+"43"+<ETX>

"00": 0x00, analog mode.

Response : <ACK>+"0A"+"D"+01"+"**00**"+"00"+"34"+<ETX>

Same as "R" command

Example 3. Tag name read/write command(21). The data format type 16S, DMFC address is 10(0x0A)

"R": Read

Sending : <STX>+"0A"+"D"+"R"+"**21**"+"64"+<ETX>

"21": Tag Name Read Command No.

Response : <ACK>+"0A"+"D"+21"+"TAG-NAME:b1a "+"00"+"1B"+<ETX>

"TAG-NAME:b1a ": 16 char. String

"W": Write

Sending : <STX>+"0A"+"D"+"W"+"**21**"+"TAG-NAME:b1a "+"6C"+<ETX>

"TAG-NAME:b1a ": Tag Name sting. The string length must be 16(with filling spaces or dots.).

Response : No response

"w": Write

Sending : Same as "W" command with different checksum.

Response : Same as "R" command

3. User commands summary

3.1 Command Table 1 : 0x00 ~ 0x1F

<table>
<tr><th>CMD</th><th>Descriptions</th><th>송수신 데이터</th><th>User 접근</th><th>B/C</th></tr>
<tr><td>0x01</td><td>Select Set-point source.
When power is applied, DMFC's default set point source is analog input signal (voltage or current). In this manual, we call it analog mode.
If set point source is digital command, we call it digital mode.
<table><tr><td>data</td><td>description</td></tr><tr><td>00</td><td>Flow controlled by analog signal (voltage or current)</td></tr><tr><td>01</td><td>Flow controlled by digital command. D</td></tr></table></td><td>2H</td><td>R/W</td><td>O</td></tr>
<tr><td>0x02</td><td>Flow on/off
After the DMFC mode is first changed to digital mode, flow rate setting with command number 09 or 0A is not applied immediately when this flow off state. User must change the DMFC at flow on state! At Flow on state, user can change the flow rate freely.

This command is for the digital mode only.
<table><tr><td>data</td><td>Description</td></tr><tr><td>00</td><td>Flow off</td></tr><tr><td>01</td><td>Flow on</td></tr></table>
c.f.
When power is on, default status is off state. This on/off status is volatile
Hint
When user want to pause the gas flow at moment, use this command instead 0x09 or 0x0A with zero flow data.</td><td>2H</td><td>R/W</td><td>O</td></tr>
<tr><td>0x03</td><td>Integration start/stop
Start and stop the flow integration.
<table><tr><td>Data</td><td>Description</td></tr><tr><td>00</td><td>W: Stop integration, R : stop status</td></tr><tr><td>01</td><td>W : start integration, R: integrating</td></tr></table>c.f.
At first time, integration time and integration amounts are start from zeros. When user restart the integration from user stopped state(ex. 1st start ➔ stop ➔ start!), previously integration time and accumulations are not</td><td>2H</td><td>R/W</td><td>O</td></tr>
</table>

<table>
<tr><td></td><td>automatically cleared.
For resetting integration time and flow amount, use command number 0x08.</td><td></td><td></td><td></td></tr>
<tr><td>0x04</td><td>Read unit flow
Read current flow rate based on DMFC/DMFM's unit.
Ex. If the flow rate is 50 and unit is SCCM, current flow rate is 50SCCM</td><td>8F/--</td><td>R</td><td>O</td></tr>
<tr><td>0x05</td><td>Read percent flow
Read current percent flow rate
Ex. If the flow rate is 50 and max flow rate of DMFC/DMFM is 200SCCM, current flow rate is 100SCCM!</td><td>8F/--</td><td>R</td><td>O</td></tr>
<tr><td>0x06</td><td>Read flow unit
Read the unit of DMFC/DMFM. Return data range is 00~1F
The units table.
<table>
<tr><th>No. (Dec.)</th><th>Units</th><th>Abbr.</th></tr>
<tr><td>00</td><td>Not yet defined. (default)</td><td></td></tr>
<tr><td>01</td><td>Standard Cubic Centimeter per Min</td><td>SCCM</td></tr>
<tr><td>02</td><td>Standard Liter per Min</td><td>SLPM</td></tr>
<tr><td>03~09</td><td>Reserved</td><td></td></tr>
<tr><td>10</td><td>Liters per second</td><td>LPS</td></tr>
<tr><td>11</td><td>Liter per minute</td><td>LPM</td></tr>
<tr><td>12</td><td>Liter per hour</td><td>LPH</td></tr>
<tr><td>13</td><td>Cubic meter per second</td><td></td></tr>
<tr><td>14</td><td>Cubic meter per minute</td><td></td></tr>
<tr><td>15</td><td>Cubic meter per hour</td><td></td></tr>
<tr><td>16</td><td>Milliliters per second</td><td></td></tr>
<tr><td>17</td><td>Milliliters per minute</td><td></td></tr>
<tr><td>18</td><td>Milliliters per hour</td><td></td></tr>
<tr><td>19</td><td>Reserved</td><td></td></tr>
<tr><td>20</td><td>Grams per second</td><td></td></tr>
<tr><td>21</td><td>Grams per minute</td><td></td></tr>
<tr><td>22</td><td>Grams per hour</td><td></td></tr>
<tr><td>23</td><td>Kilograms per second</td><td></td></tr>
<tr><td>24</td><td>Kilograms per minute</td><td></td></tr>
<tr><td>25</td><td>Kilograms per hour</td><td></td></tr>
<tr><td>26</td><td>Pounds per second</td><td></td></tr>
<tr><td>27</td><td>Pounds per minute</td><td></td></tr>
<tr><td>28</td><td>Pounds per hour</td><td></td></tr>
<tr><td>29~31</td><td>Reserved</td><td></td></tr>
</table></td><td>2H/--</td><td>R</td><td>X</td></tr>
<tr><td>0x07</td><td>Read Max Flow rate
Read the max flow rate of DMFC/DMFM.</td><td>8F/--</td><td>R</td><td>X</td></tr>
</table>

<table>
<tr><td>0x08</td><td>Read integration time and flow and clear the time and flow amounts.
Read integration time and flow amounts.
Example of responding data.
<table>
<tr><td colspan="4">Integration time (string, S)</td><td colspan="4">Integration amount (float, F)</td></tr>
<tr><td>Date</td><td>Hour</td><td>Min</td><td>Sec</td><td>LSB</td><td></td><td></td><td>MSB</td></tr>
<tr><td>99</td><td>23</td><td>59</td><td>59</td><td>0x00</td><td>0x00</td><td>0x00</td><td>0x00</td></tr>
</table>
c.f.
When writing this command without data, the integration time and flow integration is initialized to zero.
Max integration time is 99 date 23hour 59min 59sec. After this time the integration time is roll over to 0.</td><td>16SF/--
※T1</td><td>R/W</td><td>X</td></tr>
<tr><td>0x09</td><td>Unit flow set command
Flow rate setting command. If the flow rate is greater than the maximum flow rate, flow rate is automatically set to it's maximum flow rate.</td><td>8F</td><td>R/W</td><td>X</td></tr>
<tr><td>0x0A</td><td>Percent flow set command
Flow rate setting command. The flow rate is percent flow rate of it's maximum flow rate. If the flow rate is greater than the maximum flow rate, flow rate is automatically set to it's maximum flow rate.(100.0)</td><td>8F</td><td>R/W</td><td>X</td></tr>
<tr><td>0x0B</td><td>Check the kind of device
Check whether the device is DMFC or DMFM
User cannot change this value.
<table>
<tr><td>data</td><td>description</td></tr>
<tr><td>00</td><td>DMFC</td></tr>
<tr><td>04</td><td>DMFM</td></tr>
</table></td><td>2H/--</td><td>R</td><td>X</td></tr>
<tr><td>0x0C</td><td>Unit flow preset
The percent flow based flow rate preset command
This preset flow rate is only applied when the 0x0E command is sent.</td><td>8F</td><td>R/W</td><td>X</td></tr>
<tr><td>0x0D</td><td>Percent flow preset ※T3
The unit flow based flow rate preset command.
This preset flow rate is only applied when the 0x0E command is sent.</td><td>8F</td><td>R/W</td><td>X</td></tr>
<tr><td>0x0E</td><td>flow pre-set apply
This commands simultaneously apply the pre-set flow rate for the multiple DMFCs</td><td>--/2H</td><td>W</td><td>O</td></tr>
<tr><td>0x0F</td><td>Read only. Analog input output signal type(volt/current)
<table>
<tr><td>data</td><td>description</td></tr>
<tr><td>00</td><td>Voltage input and output</td></tr>
<tr><td>01</td><td>Current input and output</td></tr>
</table></td><td>2H/--</td><td>R</td><td>X</td></tr>
</table>

※T1. 16SF means 8 letter string and 8 letter hexadecimal float number.

※T2. The integration related commands are start, stop, read and initialize.

※T3. The preset function is for the simultaneous setting of multiple DMFC and multiple set points. Set each DMFC flow rate one by one with this preset command, and send command 0x0E with address FF. All flow set of connected DMFCs are simultaneously changed.

3.2 Command Table 2 : 0x20 ~ 0x2F, 0x4E~

<table>
<tr><th>CMD</th><th>Descriptions</th><th>송수신
데이터</th><th>User
접근</th><th>Broad
cast</th></tr>
<tr><td>0x20</td><td>Read the serial number of DMFC/DMFM. Read only!</td><td>16S/--</td><td>R</td><td>O</td></tr>
<tr><td>0x21</td><td>Read and write the tag Name of DMFC/DMFM.
Maximum character number is 16 and default tag name is “TAG-NAME:123456:” When user tag name is shorter than 16, user must fill the rest with appropriate characters.(ex. Spaces ‘ ’, or dots ‘..’</td><td>16S</td><td>R/W</td><td>O</td></tr>
<tr><td>0x22</td><td>Read the F/W version of DMFC/DMFM. Read only!</td><td>2H/--</td><td>R</td><td>O</td></tr>
<tr><td>0x23</td><td>Read EEPROM data format version of DMFC/DMFM. Read only!</td><td>2H/--</td><td>R</td><td>O</td></tr>
<tr><td>0x24</td><td>Read the board version of DMFC/DMFM. Read only!</td><td>2H/--</td><td>R</td><td>O</td></tr>
<tr><td>0x25</td><td>Read Assemble version of DMFC/DMFM. Read only!</td><td>2H/--</td><td>R</td><td>O</td></tr>
<tr><td>0x26</td><td>Read all the PCBs information at once.
A command for the 0x22~0x25 command.
Example of Response <ACK> 13D26442251100047<EOT>
<table>
<tr><th>Gas</th><th>Number</th></tr>
<tr><td>44</td><td>F/W version 4.4</td></tr>
<tr><td>22</td><td>EERPRM format 2.2</td></tr>
<tr><td>51</td><td>PCB version 5.1</td></tr>
<tr><td>10</td><td>Assemble 1.0</td></tr>
</table></td><td>8H</td><td>R</td><td>O</td></tr>
<tr><td>0x28</td><td>Read and write the address of DMFC/DMFM
The range of address is from 0x01 to 0xFE. The address 0x00 is assigned for the PC and 0xFF is assigned for the broadcast.</td><td>2H</td><td>R/W</td><td>O</td></tr>
<tr><td>0x29</td><td>Read and write(set) Baud rate of DMFC/DMFM
Read and write command for baud rate of DMFC/DMFM
The changed baud rate is applied after reset the reset of power
<table>
<tr><th>Data</th><th>Baud rate</th></tr>
<tr><td>00</td><td>9600 bps</td></tr>
<tr><td>01</td><td>19200 bps</td></tr>
<tr><td>02</td><td>38400 bps</td></tr>
<tr><td>03</td><td>57600 bps (factory default)</td></tr>
<tr><td>04</td><td>115200 bps</td></tr>
<tr><td>05~</td><td>Ignore</td></tr>
</table></td><td>2H</td><td>R/W</td><td>O</td></tr>
<tr><td>0x2A</td><td>Gas Number</td><td>2H</td><td>R/W</td><td></td></tr>
</table>

<table>
<tr><td></td><td>DMFC/DMFM gas number. Refer Appendix
<table><tr><td>Data (index)</td><td>Gas name</td></tr><tr><td>0</td><td>User</td></tr><tr><td>1</td><td>Acetylene</td></tr><tr><td>2</td><td>Air</td></tr><tr><td>3</td><td>Allene</td></tr><tr><td>4</td><td>…</td></tr></table>
See the appendix for the complete list of gas number and it’s gas name.</td><td></td><td></td><td></td></tr>
<tr><td>0x2B</td><td>Read the gas factor and set the user gas factor.

See the appendix for the complete list of gas factor.
<table><tr><td>Index</td><td>Gas name</td><td>Gas factor</td></tr><tr><td>0</td><td>User</td><td>User input</td></tr><tr><td>1</td><td>Acetylene</td><td>0.615</td></tr><tr><td>2</td><td>Air</td><td>0.998</td></tr><tr><td>3</td><td>Allene</td><td>0.478</td></tr><tr><td>4</td><td>…</td><td>..</td></tr></table>
When the current gas number is 0, user must set the user gas factor.
With “W’ command user set the gas factor in the range of 0.1 to 4.0..
Default user gas factor is 1.0.</td><td>8F</td><td>R/W</td><td></td></tr>
<tr><td>0x2C</td><td>Read/write gas formula
For the user’s convenience, user can change this gas formula.</td><td>16S</td><td>R/W</td><td></td></tr>
<tr><td>0x2D</td><td>Read VOR status
Read the DMFC VOR(valve override) status

<table><tr><td>No</td><td>Descriptions.</td></tr><tr><td>00</td><td>Normal operation</td></tr><tr><td>01</td><td>Low signal input at VOR pin. Valve closed/</td></tr><tr><td>02</td><td>High signal input at VOR pin. Valve fully opened.</td></tr></table></td><td>2H</td><td>R</td><td></td></tr>
<tr><td>0x2E</td><td>Reserved</td><td></td><td></td><td></td></tr>
<tr><td>0x2F</td><td>Percent flow and percent PID value
Read current percent flow rate and percent PWM value(proportional to solenoid valve voltage) at once.
The nominal range of two values is 0~100.0%
This command is not recommended general use.</td><td>8F8F/-</td><td>R</td><td></td></tr>
</table>

Appendix 1

ASCII Table :: 0x00~0x7F

Hex	Dec	Char	Hex	Dec	Char	Hex	Dec	Char	Hex	Dec	Char
0x00	0	NULL	0x20	33		0x40	65	@	0x60	97	`
0x01	1	SOH	0x21	34	!	0x41	66	A	0x61	98	a
0x02	2	STX	0x22	35	=	0x42	67	B	0x62	99	b
0x03	3	ETX	0x23	36	#	0x43	68	C	0x63	100	c
0x04	4	EOT	0x24	37	$	0x44	69	D	0x64	101	d
0x05	5	ENQ	0x25	38	%	0x45	70	E	0x65	102	e
0x06	6	ACK	0x26	39	&	0x46	71	F	0x66	103	f
0x07	7	BELL	0x27	40	'	0x47	72	G	0x67	104	g
0x08	8	BS	0x28	41	(	0x48	73	H	0x68	105	h
0x09	9	TAB	0x29	42	)	0x49	74	I	0x69	106	i
0x0A	10	LF	0x2A	43	*	0x4A	75	J	0x6A	107	j
0x0B	11	VT	0x2B	44	+	0x4B	76	K	0x6B	108	k
0x0C	12	FF	0x2C	45	'	0x4C	77	L	0x6C	109	l
0x0D	13	CR	0x2D	46	-	0x4D	78	M	0x6D	110	m
0x0E	14	SO	0x2E	47	.	0x4E	79	N	0x6E	111	n
0x0F	15	SI	0x2F	48	/	0x4F	80	O	0x6F	112	o
0x00	16	DLE	0x30	49	0	0x50	81	P	0x70	113	p
0x01	19	DC1	0x31	50	1	0x51	82	Q	0x71	114	q
0x02	19	DC2	0x32	51	2	0x52	83	R	0x72	115	r
0x03	20	DC3	0x33	52	3	0x53	84	S	0x73	116	s
0x04	21	DC4	0x34	53	4	0x54	85	T	0x74	117	t
0x05	22	NAK	0x35	54	5	0x55	86	U	0x75	118	u
0x06	23	SYN	0x36	55	6	0x56	87	V	0x76	119	v
0x07	24	ETB	0x37	56	7	0x57	88	W	0x77	120	w
0x08	25	CAN	0x38	57	8	0x58	89	X	0x78	121	x
0x09	26	EM	0x39	58	9	0x59	90	Y	0x79	122	y
0x0A	27	SUB	0x3A	59	:	0x5A	91	Z	0x7A	123	z
0x0B	28	FSC	0x3B	60	;	0x5B	92	[	0x7B	124	{
0x0C	29	FS	0x3C	61	<	0x5C	93	\	0x7C	125	|
0x0D	30	GS	0x3D	62	=	0x5D	94	]	0x7D	126	|
0x0E	31	RS	0x3E	63	>	0x5E	95	^	0x7E	127	~
0x0F	32	US	0x3F	64	?	0x5F	96	_	0x7F	128	DEL

Hexadecimal, binary, decimal numbers notation and conversion table

Decimal	Hex.	Bin.	Decimal	Hex.	Bin.
0	0	0000	8	8	1000
1	1	0001	9	9	1001
2	2	0010	10	A	1010
3	3	0011	11	B	1011
4	4	0100	12	C	1100
5	5	0101	13	D	1101
6	6	0110	14	E	1110
7	7	0111	15	F	1111

Appendix 2. Tables

Table A2-1. Unit lists

No	Units	Abbr.
00	Not yet defined. (default)	
01	Standard Cubic Centimeter per Min	SCCM
02	Standard Liter per Min	SLPM
03~09	Reserved	
10	Liters per second	LPS
11	Liter per minute	LPM
12	Liter per hour	LPH
13	Cubic meter per second	
14	Cubic meter per minute	
15	Cubic meter per hour	
16	Milliliters per second	
17	Milliliters per minute	
18	Milliliters per hour	
19	Reserved	
20	Grams per second	
21	Grams per minute	
22	Grams per hour	
23	Kilograms per second	
24	Kilograms per minute	
25	Kilograms per hour	
26	Pounds per second	
27	Pounds per minute	
28	Pounds per hour	
29~31	Reserved	

C.f. ::

Unit number 00 means "The flow unit is not yet determined!"

A2-2. Restrictor

No.	Restrictor
0	Not yet assigned
1	PL
2	E
3	F
4	G
5	H
6	J
7	K
8	M
9	N
10	P
11	Q
12	R
13	T
14	U
15	V
16	W
17	XY
18	1
19	3
20	4
21	5
22	301
23	302
24	303
25	401
26	304
27	405
28	602/403
29	L1
30	M1
31	H1

Appendix 3. Gas number, name, factor table.

No	gas name	gas	G. F.
0	Mixture/User	User	1.000
1	Acetylene	C2H2	0.615
2	Air	Air	0.998
3	Allene	C3H4	0.478
4	Ammonia	NH3	0.786
5	Argon	Ar	1.395
6	Arsine	AsH3	0.754
7	Boron Trichloride	BCL3	0.443
8	Boron Trifluoride	BF3	0.579
9	Bromine Pentafluoride	BrF5	0.287
10	Bromine Trifluoride	BrF3	0.439
11	Bromotrifluoroethylene	C2BrF3	0.326
12	Bromotrifluoromethane f-13B1	CBrF3	0.412
13	1,3-Butadiene	C4H6	0.354
14	Butane	C4H10	0.257
15	1-Butene	C4H8	0.294
16	CIS-2-Butene	C4H8	0.320
17	Trans-2-Butene	C4H8	0.291
18	Carbon Dioxide	CO2	0.740
19	Carbon Disulfide	CS2	0.638
20	Carbon Monoxide	CO	0.995
21	Carbon Tetrachloride	CCL4	0.344
22	Carbon Tetrafluoride f-14	CF4	0.440
23	Carbonyl Fluoride	COF2	0.567
24	Carbonyl Sulfide	COS	0.680
25	Chlorine	CL2	0.876
26	Chlorine Dioxide	CLO2	0.693
27	Chlorine Trifluoride	CLF3	0.433
28	2-Chlorobutane	C4H9Cl	0.234
29	Chlorodifluoromethane f-22	CHCLF2	0.505
30	Chloroform (Trichloromethane)	CHCL3	0.442
31	Chloropentafluoroethane f-115	C2CLF5	0.243
32	Chlorotrifluoroethylene	C2CLF3	0.337
33	Chlorotrifluoromethane f-13	CCLF3	0.430
34	Cyanogen	(CN)2	0.498
35	Cyanogen Chloride	CLCN	0.618
36	Cyclobutane	C4H8	0.387

37	Cyclopropane	C3H6	0.505
38	Deuterium	D2	0.995
39	Diborane	B2H6	0.448
40	Diboromodifluoromethane f-12B2	CBr2F2	0.363
41	1,2-Dibromotetrafluoroethane f-114B2	C2Br2F4	0.215
42	Dichlorodifluoromethane f-12	CCL2F2	0.390
43	Dichlorofluoromethane f-21	CHCL2F	0.456
44	Dichlorosilane	SiH2CL2	0.442
45	1,2-Dichloroethane	C2H4Cl2	0.382
46	1,2-Dichlorotetrafluoroethane f-114	C2CL2F4	0.231
47	2,2 Dichloro	C2HC12F3	0.259
48	1,1-Difluoro-1-Chloroethane	C2H3CLF2	0.341
49	1,1-Difluoroethane	CH3CHF2	0.415
50	1,1-Difluoroethylene	CH2CF2	0.458
51	Diedthylsilane	C4H12Si	0.183
52	Difluoromethane f-32	CF2H2	0.627
53	Dimethylamine	(CH3)2NH	0.370
54	Dimethylether	(CH3)2O	0.392
55	2,2-Dimethylpropane	C(CH3)4	0.247
56	Disilane	Si2H6	0.332
57	Ethane	C2H6	0.490
58	Ethanol	C2H6O	0.394
59	Ethylacetylene	C4H6	0.365
60	Ethyl Chloride	C2H5CL	0.408
61	Ethylene	C2H4	0.619
62	Ethylene Oxide	C2H4O	0.589
63	Fluorine	F2	0.924
64	Fluoroform f-23	CHF3	0.529
65	Germane	GeH4	0.649
66	Germanium Tetrachloride	GeCl4	0.268
67	Halothane (R-123B1)	C2HBrCLF3	0.257
68	Helium	He	1.386
69	Hexafluoroacetone	F3CCOCF3	0.219
70	Hexafluorobenzine	C6F6	0.632
71	Hexafluoroethane f-116	C2F6	0.255
72	Hexafluoropropylene (HFP)	C3F6	0.249
73	Hexamethyldisilane (HMDS)	(CH2)6Si2	0.139
74	Hexane	C6H14	0.204
75	Hydrogen	H2	1.008
76	Hydrogen Bromide	HBr	0.987

77	Hydrogen Chloride	HCL	0.983
78	Hydrogen Cyanide	HCN	0.744
79	Hydrogen Iodide	HF	0.998
80	Hydrogen Iodide	HI	0.953
81	Hydrogen Selenide	H2Se	0.837
82	Hydrogen Sulfide	H2S	0.850
83	Iodine Pentafluoride	IF5	0.283
84	Isobutane	C4H10	0.260
85	Isobutene	C4H8	0.289
86	Isopentane	C5H12	0.211
87	Krypton	Kr	1.382
88	Methane	CH4	0.763
89	Methylacetylene	C3H4	0.473
90	Methyl Bromide	CH3Br	0.646
91	3-Methyl-1-butene	C5H10	0.252
92	Methyl Chloride	CH3CL	0.687
93	Methyl Fluoride	CH3F	0.761
94	Methyl Mercaptan	CH4S	0.588
95	Methyl Silane	C3H6Si	0.393
96	Methyl Trichlorosilane (MTS)	CH3Cl3Si	0.267
97	Methyl Vinyl Ether	C3H6O	0.377
98	Monoethyanolamine	C2H7NO	0.305
99	Monoethylamine	C2HH5NH2	0.359
100	Monomethylamine	CH3NH2	0.565
101	Neon	Ne	1.398
102	Nickel Carbonyl	Ni(CO)4	0.212
103	Nitric Oxide	NO	0.995
104	Nitrogen	N2	1.000
105	Nitrogen Dioxide	NO2	0.758
106	Nitrogen Trifluoride	NF3	0.501
107	Nitrogen Trioxide	N2O3	0.443
108	Nitrosyl Chloride	NOCL	0.644
109	Nitrous Oxide	N2O	0.752
110	Octofluorocyclobutane	C4F8	0.169
111	Oxygen	O2	0.988
112	Oxygen Difluoride	OF2	0.672
113	Ozone	O3	0.738
114	Pentafluoroethane f-125	C2HF5	0.287
115	Pentane (n-Pentane)	C5H12	0.212
116	Perchloryl Fluoride	CLO3F	0.448

117	Perfluorobutane	C4F10	0.738
118	Perfluoro-2-Butene	C4F8	0.268
119	Perfluoromethyl-vinylether	PMVE	0.296
120	Perfluoropropane	C3F8	0.179
121	Phosgene	COCL2	0.504
122	Phosphine	PH3	0.783
123	Phosphorous Pentafluoride	PF5	0.346
124	Phosphorous Trifluoridide	PF3	0.495
125	Propane (same as CH3CH2CH3)	C3H8	0.343
126	Propylene (Propene)	C3H6	0.401
127	Rhenium Hexafluoride	ReF6	0.230
128	Silane	SiH4	0.625
129	Silicon Tetrachloride	SiCL4	0.310
130	Silicon Tetrafluoride	SiF4	0.395
131	Sulfur Dioxide	SO2	0.728
132	Sulfur Hexafluoride	SF6	0.270
133	Sulfur Tetrafluoride	SF4	0.353
134	Sulfur Trioxide	SO3	0.535
135	Sulfuryl Fluoride	SO2F2	0.423
136	Tetrachloromethane	CCL4	0.344
137	Tetrafluoroethylene (TFE)	C2F4	0.361
138	Tetrafluorohydrazine	N2F4	0.367
139	Trichlorofluoromethane f-11	CCL3F	0.374
140	Trichlorosilane	SiHCL3	0.329
141	Trimethyloxyborane (TMB)	B(OCH3)3	0.300
142	1,1,2-Trichloro-1,1,2-Trifluroet	C2CL3F3	0.231
143	Trimethylamine	(CH3)3N	0.316
144	Tungsten Hexafluoride	WF6	0.227
145	Uranium Hexafluoride	UF6	0.220
146	Vinyl Bromide	C2H3Br	0.524
147	Vinyl Chloride	C2H3CL	0.542
148	Vinyl Fluoride	C2H3F	0.576
149	Water Vapor	H2O	0.861
150	Xenon	Xe	1.383